# DR. MADHAB BARMAN

Postdoctoral Fellow @ IIM–Bangalore

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## CARRIER OBJECTIVE

I am passionate about using my expertise in data-driven spatial extreme value theory, network-driven analysis, epidemic-informed machine learning for spatiotemporal analysis, stochastic energy balanced models, and computational modeling to make a meaningful impact in modern research. By collaborating with diverse teams and building advanced methods, I aim to leverage my research in statistical informatics to provide data-driven insights for public health and environmental risk, contributing to cutting-edge research and collaborative innovation in modern decision sciences.

## RESEARCH INTERESTS

Network-Driven Dynamics, Stability of Dynamical Systems, Bifurcation Analysis, Random ordinary differential equations, Stochastic Perturbations, Infectious Disease Modeling, Epidemic-Informed Deep Learning, Optimization in Machine Learning, Graph Neural Networks, and Health Data Science.

My current research centers on data-driven estimation of rare event probabilities using Spatial Extreme Value Theory (SEVT), Stochastic Energy Balance Climate Modeling, and Physics-Informed Neural Networks (PINNs) for Disease Modeling.

## WORK EXPERIENCE

**Indian Institute Of Management–Bangalore (IIM–Bangalore), India** *June 2025 - Present*
Postdoctoral Fellow, Decision Sciences
(*Data-Driven Estimation of Rare Event Probabilities for Spatial Extremes*)

**Sorbonne University, Abu Dhabi, UAE** *Oct 2024 - May 2025*
Visiting Researcher, ForeML Lab (8 Months)
Department Of Mathematics (Statistics & Data Science)
(*Deep Learning in Networked-Spatial Modeling and Dynamical Systems*)

## EDUCATION

**IIITDM Kancheepuram, Chennai, Tamil Nadu** *2018- 2024*
Doctor of Philosophy (Ph.D.), CAM Lab
Department Of Mathematics
(*Analysis and Simulations of Network-Based Nonlinear Epidemic Models*)

**Visva-Bharati, Santiniketan, West Bengal** *2016 - 2018*
Master of Science in Mathematics — Overall Percentage: 88.8
*(Dissertation: Dynamics of species in a two predators-one prey model)*

**A.B.N Seal College, North Bengal University, West Bengal** *2013 -2016*
Bachelor of Science in Mathematics — Overall Percentage: 71.75

## ACADEMIC ACHIEVEMENTS

1. **Best paper award** at the 4th Conference on Information and Communication Technology (CICT-2020).
2. Institute fellowship for Half Teaching Research Assistant (**HTRA**) during PhD.
3. INSPIRE **Scholarship** for undergraduate/masters studies (**SHE**)
4. INSPIRE **Fellowship** for PhD (awarded).
5. **GATE** Entrance Examination qualified in 2017.

6. **CSIR NET** Examination qualified in 2017.
7. **Sitesh Ch. Chattopadhyay Mem. Prize** for achieving the highest marks in B.Sc 2nd year.

## RESEARCH PUBLICATIONS

### Published Articles

1. **M. Barman** and N. Mishra, Hopf Bifurcation in Networked Delay SIR Epidemic Model, Journal of Mathematical Analysis and Applications, 2023, 127131, `10.1016/j.jmaa.2023.127131`.
2. **M. Barman** and N. Mishra, Hopf Bifurcation and Its Direction in Networked Delay SEIR Epidemic Model, Chaos, Solitons & Fractals, 178(2024)114351. DOI: `10.1016/j.chaos.2023.114351`.
3. **M. Barman** and N. Mishra, Network-Driven Global Analysis for an SVIRS Epidemic Model, Journal of Biological Systems, 2024, DOI: `10.1142/S0218339025500093`
4. **M. Barman** and N. Mishra, Network-driven Stability Analysis and Optimal Control for a Weighted Networked SAIR Model in Epidemiology, International Journal of Dynamics and Control, 13(53), 2025, DOI: 10.1007/s40435-024-01556-8
5. **M. Barman**, M. Panja, N. Mishra, T. Chakraborty, Epidemic-guided deep learning for spatiotemporal forecasting of Tuberculosis outbreak, Machine Learning, (2025) 114:213, `10.1007/s10994-025-06848-4`.

### Published Articles in Conferences

- **M. Barman** and N. Mishra, A time-delay SEAIR model for COVID-19 spread, IEEE Conference Proceedings; 2020, DOI: 10.1109/CICT51604.2020.9312111 (**Best Paper Award**).
- **M. Barman** and N. Mishra, Susceptible-Infected-Removed Epidemic Model On Network: A Migration Flow Study, 2022 IEEE Conference on Information and Communication Technology, Gwalior, India, 2022, pp. 1-6, DOI: `10.1109/CICT56698.2022.9997851`.

### Submitted for Publication

1. **M. Barman** and N. Mishra, A Caputo-based Fractional-order SEIR Epidemic Model on Weighted Networks, 2026+ (Submitted to Applied Mathematical Modelling).

### Work in Progres (in preparation)

1. **Madhab Barman**, Lizan Pereira, Anand Deo, Soudeep Deb, Data-driven estimation of rare event probabilities for spatial extremes with applications to rainfall data .
2. **Madhab Barman**, Indrajit Ghosh, Tanujit Chakraborty, Spatiotemporal dynamics and intervention assessment for influenza-like illness using a networked time series SIR model.
3. Prince Achankunju, **Madhab Barman**, Nachiketa Mishra, and Saroj Kumar Dash, Graph Laplacian-Driven SEIR Epidemic Model with Stochastic Perturbations in Transmission Rate: Stability and Spatiotemporal Analysis.

## TECHNICAL PROFICIENCY

**Software and Programming Skills (Advanced Level)** Python, R, Matlab, Julia, Latex, and C.

## ACADEMIC PROJECT

**M.Sc. Dissertation (Specialization: Bio-Mathematics)**
**Title:** Dynamics of species in a two predators-one prey model
**Summery:** In this project, a mathematical model of two competing predators sharing one prey is proposed and investigated. The motivation is to answer the questions of the persistence of the system and extinction of one the predator based on the efficient conversion of prey biomass. It is explained analytically and numerically in the case of a non-periodic solution.

## ACADEMIC ACTIVITIES

1. Optimization Techniques in Machine Learning- Teaching Assistant for B.Tech.
2. Computer Lab- Teaching Assistant for B.Tech.

3. Numerical Methods- Teaching Assistant for B.Tech.
4. Linear Algebra- Teaching Assistant for B.Tech.
5. Calculus- Teaching Assistant for B.Tech.
6. Ordinary Differential Equations (ODE)- Teaching Assistant for B.Tech.
7. Reviewer:
    - Journal of Applied Mathematics and Computing
    - International Journal of Dynamics and Control
    - Journal of Applied Analysis and Computation
    - Network Modeling Analysis in Health Informatics and Bioinformatics
    - Journal of Applied Nonlinear Dynamics
    - Zeitschrift für angewandte Mathematik und Physik

## PRESENTATIONS

1. **Departmental Seminars/Symposia/Talks**
    - Epidemic Modeling and Simulations, Sorbonne University, Abu Dhabi, 2024
    - PhD Seminar-I & Seminar-II, IIITDM Kancheepuram, Chennai, 2024
    - Decision Sciences Area Brown Bag Seminar, IIM Bangalore, Bengaluru, India, 2025.
2. **At Conferences / Specialized Workshops / Courses**
    - Data-Driven Estimation of Rare Event Probabilities for Spatial Extremes and Its Applications, **International Symposium on Nonparametric Statistics (ISNPS)**, 22-26 June, 2026, Thessaloniki, Greece.
    - Data-Driven Estimation of Rare Event Probabilities for Spatial Extremes, **International Workshop on Advances in Risk Analysis & Management: A focus on Emergent Risks**, Singapore Campus ESSEC-Asia-Pacific, February 26-27, 2026, Singapore.
    - Migration-based Spatial Network in Application Epidemiology, **MMTRC** @ IIITDM Kancheepuram, Chennai, 2025
    - Abstract accepted (Submission #589) for the **Third International Nonlinear Dynamics Conference (NODYCON)**, June 18-22, 2023, Rome, Italy.

## DISCUSSION MEETING/WORKSHOP/CONFERENCE

1. **The 17th Annual Edition of the IMR Doctoral Conference**, 9 January 2026 to 10 January 2026, Indian Institute of Management Bangalore.
2. **ACM India Winter School 2025: AI and Finance** (Official Pre-Summit Event of the AI Impact Summit 2026), 8 December 2025 to 13 December 2025, IIIT Hyderabad, Hyderabad.
3. **An Online Workshop on Fundamentals of Mathematical Biology,** 11 May - 12 May 2023; Bioinformatics Centre, Indian Institute of Information Technology Allahabad, India.
4. **Lecture Series on Nonlinear Dynamics and Applications,** 13 Feb - 16 Feb, 2023; Indian Institute of Technology Indore, India.
5. **EWM-EMS Summer School on Multi-scale Modeling for Pattern Formation in Biological Systems,** July 19 - July 23, 2021; Institut Mittag-Leffler, Sweden [**Online**].
6. **Multi-Scale Analysis: Thematic Lectures and Meeting (MATHLEC-2021),** 15 February 2021 to 19 February 2021; ICTS, Bangalore [**Online**].
7. **Workshop on Finite Elements for Nonlinear and Multiscale Problems,** Feb 28- Mar 3, 2020, Department of Mathematics, IISc Bangalore.

## REFERENCES

1. **Dr. Nachiketa Mishra**
   Assistant Professor of Department of Sciences and Humanities (Mathematics)
   Indian Institute of Information Technology, Design and Manufacturing, Kancheepuram, Chennai, India.
   **Email:** nmishra@iiitdm.ac.in
   **Homepage:** https://www.iiitdm.ac.in/people/faculty/nmishra@iiitdm.ac.in

2. **Dr. Soudeep Deb**
   Associate Professor of Decision Sciences (Chairperson)
   Indian Institute Of Management–Bangalore, Bilekahalli, Bengaluru, Karnataka 560076.
   **Email:** soudeep@iimb.ac.in
   **Homepage:** https://www.iimb.ac.in/user/196/soudeep-deb

3. **Dr. Anand Deo**
   Assistant Professor of Decision Sciences
   Indian Institute Of Management–Bangalore, Bilekahalli, Bengaluru, Karnataka 560076.
   **Email:** anand.deo@iimb.ac.in
   **Homepage:** https://www.iimb.ac.in/user/244/anand-deo